Laila C. Johnston

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Education

currently **Ph.D. in Cognitive Science**
Brown University, Providence, RI, USA
Advisors: Roman Feiman & Ellie Pavlick

2026 **M.Sc. in Cognitive Science**
Brown University, Providence, RI, USA
Advisors: Roman Feiman & Ellie Pavlick

2023 **B.S. in Mathematics**
Minors in Computer Science and Philosophy
University of Central Florida, Orlando, FL, USA

Research Interests

computational cognitive science ~ compositional concepts ~ concept learning ~ hypothesis spaces ~ natural language progressing ~ representation spaces ~ the frame problem ~ pragmatics ~ ad-hoc alternatives ~ the relevance problem

Research Experience

2023 – present **Ph.D. Student**, *Language and Thought Lab*, Brown University, Providence, RI
Advisor: Prof. Roman Feiman

2023 – present **Ph.D. Student**, *LUNAR Lab*, Brown University, Providence, RI
Advisor: Prof. Ellie Pavlick

2021 – 2023 ***CoCoSci Group***, Massachusetts Institute of Technology, Cambridge, MA
Undergraduate Researcher (Sep. 2022 – May 2023)
Research Fellow, MSRP Bio/Neuro (Summer 2022)
Visiting Student (Aug. 2021 – Dec. 2021)
Advisor: Prof. Joshua B. Tenenbaum

Summer 2021 **Research Fellow**, MSRP Bio/Neuro, Center for Brains, Minds, and Machines, *Computational Cognitive Neuroscience Lab*, Harvard University, Cambridge, MA
Advisor: Prof. Samuel J. Gershman

2020 – 2021 ***Center for the Neural Basis of Cognition***, Carnegie Mellon University, Pittsburgh, PA
Undergraduate Researcher (Sep. 2020 – Feb. 2021)
Research Fellow, Undergraduate Program in Neural Computation (Summer 2020)
Advisor: Prof. David Danks

Awards and Honors

2023 **National Science Foundation Graduate Research Fellowship**
Feb. 2022 **Mathematics of Collective Intelligence Workshop Travel Scholarship**, IPAM, UCLA
Fall 2021 **Visiting Student Fellowship**, Department of Brain and Cognitive Sciences, MIT
2018 – 2021 **Dean's List** (5 Semesters), University of Central Florida
2021 **Hispanic Heritage Scholarship Fund of Metro Orlando Scholar**
2020 **Ronald E. McNair Scholar**

2020 **Carolyn Euliano Endowed Scholarship in Mathematics**, University of Central Florida
2018 **EXCEL Scholar**, University of Central Florida
2018 **Pegasus Scholar**, University of Central Florida
2018 **Florida's Bright Futures Academic Scholar**
2018 **International Baccalaureate Diploma Recipient**

Publications

Michael A. Lepori, **Laila Johnston**, Aalok Sathe, Zhuonan Yang, Roman Feiman, Ellie Pavlick. (2026). An Algorithmic Account of Few-Shot Concept Induction in Language Models. ***Findings of the Association for Computational Linguistics: EMNLP.***

Laila Johnston*, Noah Hillman*, David Danks. (2021). Individual Differences in Causal Learning. ***Proceedings of the 43rd Annual Conference of the Cognitive Science Society.***

** co-author*

Posters and Abstracts

Laila Johnston, Daniel A. Smits, Jason Ritt, Ellie Pavlivk, Roman Feiman. (2026, March). In-Context Next-Word Prediction as a Mechanism for Generating Focus Alternatives. ***Human Sentence Processing.*** Abstract accepted.

Laila Johnston [Δ], Daniel A. Smits, Ellie Pavlivk, Roman Feiman. (2024, June). The Structure of Ad-Hoc Alternatives. ***Experiments in Linguistic Meaning.*** Poster presentation.

Laila Johnston [Δ], Max H. Siegel, Joshua B. Tenenbaum, Tobias Gerstenberg. (2022, October). Reasoning with Compositional Concepts in the Probabilistic Language of Thought. ***SACNAS NDiSTEM Research Conference.*** Poster presentation.

Laila Johnston [Δ], Max H. Siegel, Joshua B. Tenenbaum, Tobias Gerstenberg. (2022, August). Reasoning with Compositional Concepts in the Probabilistic Language of Thought. ***Center for Brains, Minds, and Machines Summer Research Poster Session.*** Poster presentation.

Laila Johnston [Δ], Max H. Siegel, Joshua B. Tenenbaum, Tobias Gerstenberg. (2021, September). Reasoning with Compositional Concepts. ***MKN McNair Heartland Research Conference.*** Oral presentation (15 minutes).

Laila Johnston [Δ], Christopher J. Bates, B. Egger, Samuel J. Gershman. (2021, August). Scaling Models of Visual Working Memory to Natural Images: A Case Study in Human Faces. ***Center for Brains, Minds, and Machines Summer Research Poster Session.*** Poster presentation.

Laila Johnston [Δ], Noah Hillman, David Danks. (2021, March). Individual Differences in Causal Learning. ***UCF Student Scholar Symposium.*** Poster presentation.

Laila Johnston [Δ], Noah Hillman [Δ], David Danks. (2020, August). Individual Variation in Causal Learning. ***Center for the Neural Basis of Cognition Undergraduate Summer Research Showcase.*** Poster presentation. Video presentation.

[Δ] presenter